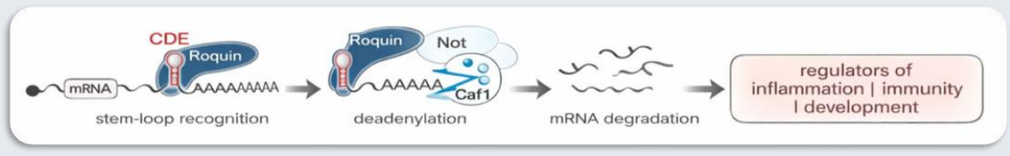
Contributions of mRNA Sequence and Secondary Structure to Roquin-1 Binding Specificity

Aviram Siginur, Amit Rendlich
Introduction to Bioinformatics – 136158

Introduction

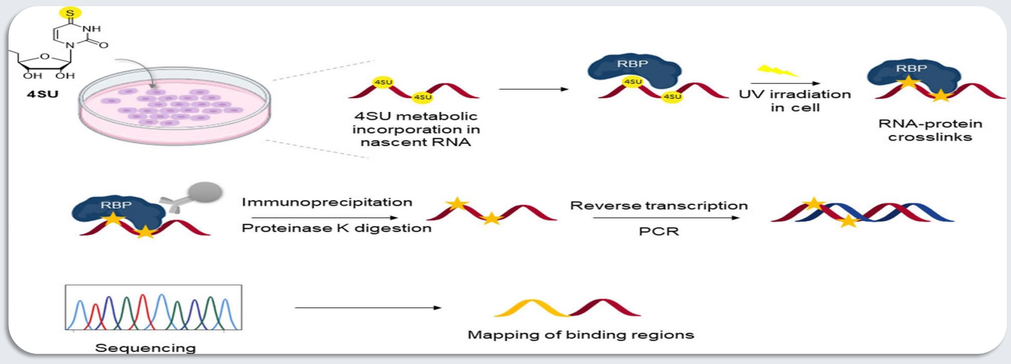
Roquin-1 (RC3H1) is an RNA-binding protein that regulates mRNA stability by binding motifs in the 3'UTR of target transcripts involved in immune and inflammatory responses. Its binding is classically associated with the constitutive decay element (CDE), a stem - loop RNA structure. However, many binding sites lack a canonical CDE (UCYRYGA) and instead contain U-rich sequences within hairpin structures, suggesting that recognition may depend on RNA structure rather than sequence alone.

In this project, we analyzed PAR-CLIP data to identify enriched motifs at RC3H1 RNA targets and examine their structural context.



Dataset

- PAR-CLIP dataset from human HEK293 cells expressing Roquin-1 (RC3H1)
- ~3,800 target mRNAs and >16,000 binding sites
- Dataset accession: GEO GSM1144507

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chr18   55722000        55722060        CID_006127      1000    +       55722038        55722039
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Main Tools

RStudio — Data analysis and visualization

UCSC Genome Browser — A comprehensive database and browser for genomic annotations

DRIMust — Discovery of enriched RNA sequence motifs

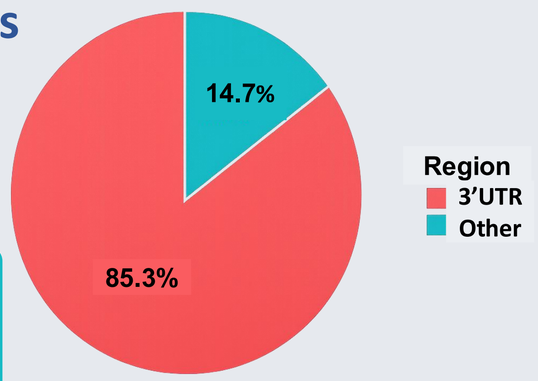
ViennaRNA Web Services (Institute for Theoretical Chemistry) — Prediction of RNA secondary structure

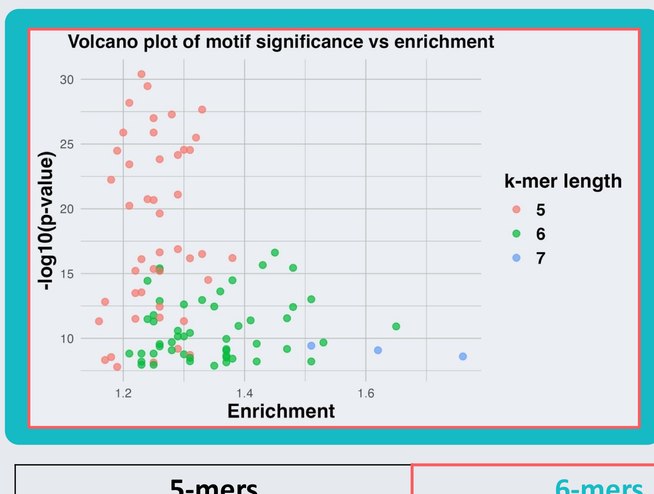
RBPmap (Mapping Binding Sites of RNA Binding Proteins) — Identification of RNA-binding protein binding sites

RCSB PDB (PROTEIN DATA BANK) — An archive for 3D structural data of large biological molecules

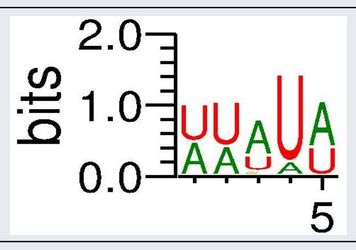
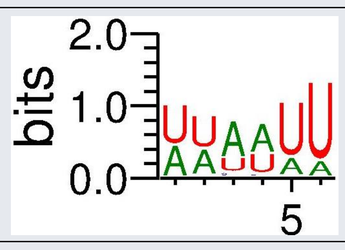
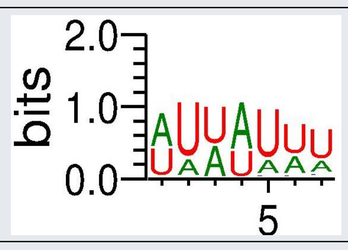
Results

1. The PAR-CLIP dataset was processed and RC3H1 binding sites were extracted and classified by genomic region.

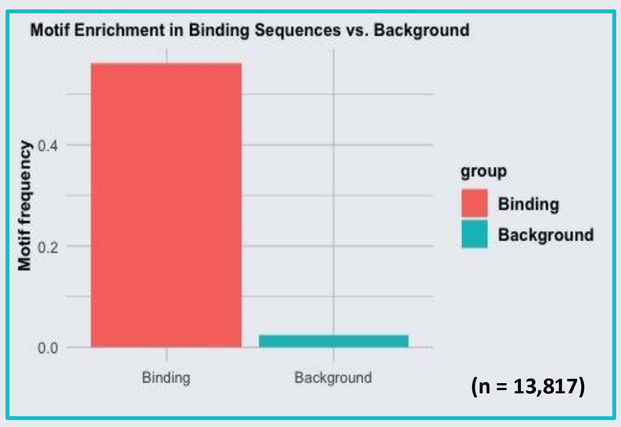




2. The extracted binding-site sequences were converted into FASTA format and analyzed using **DRIMust** to determine the optimal motif length and degenerate sequence logos.

5-mers	6-mers	7-mers
		
$P-value \leq 1.3 \times 10^{-136}$	$P-value \leq 1.3 \times 10^{-156}$	$P-value \leq 3.8 \times 10^{-120}$

3. Sequences containing degenerate motifs from the 6-mers sequence logo were analyzed using **RBPmap** confirmed that motifs identified by DRIMust remain significantly enriched in binding sequences compared to shuffled background versions of the same sequences.

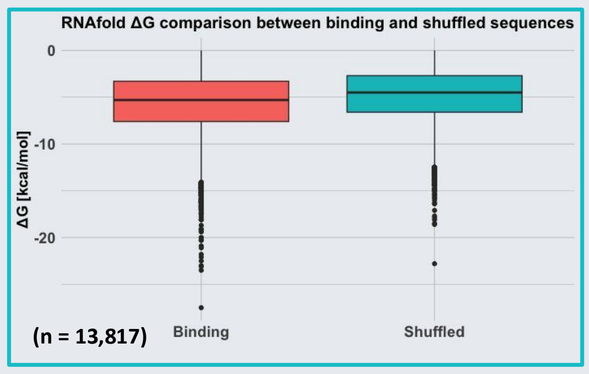


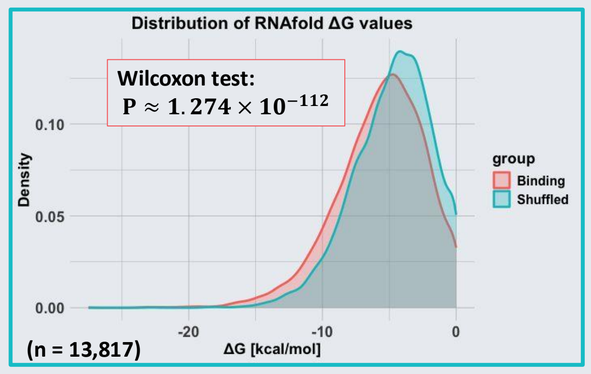
Fisher exact test	$P \leq 2.2 \times 10^{-16}$
Enrichment	$\frac{(M/N)_{binding}}{(M/N)_{background}} \approx 22.5$

$M_{binding/background}$ – number of motif occurrences in binding/background sequences.

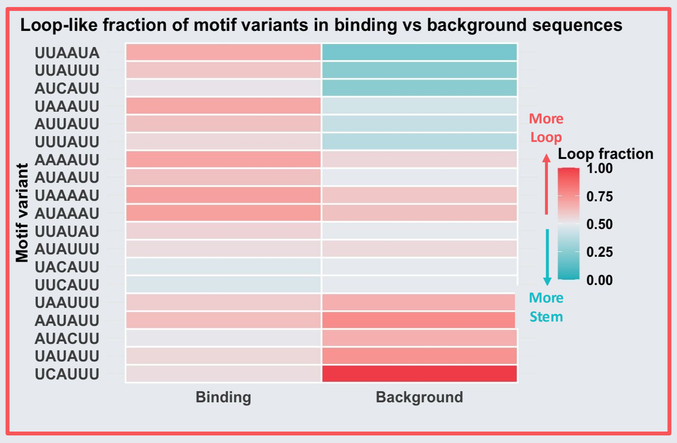
$N_{binding/background}$ – total number of binding/background sequences.

4. Binding sequences and their shuffled background were analyzed with **ViennaRNA (RNAfold)** to calculate ΔG and assess structural enrichment.





5. To investigate whether the enriched motif preferentially occurs in specific RNA structural contexts, the fraction of loop-like occurrences was calculated for each exact motif variant similarly to previous comparison.



Fisher exact test	$P \approx 0.015$
Loop Enrichment	$\frac{(M/N)^{binding}_{Loop}}{(M/N)^{background}_{Loop}} \approx 1.133$

M_{Loop} – number of motif occurrences in loop binding/background sequences.

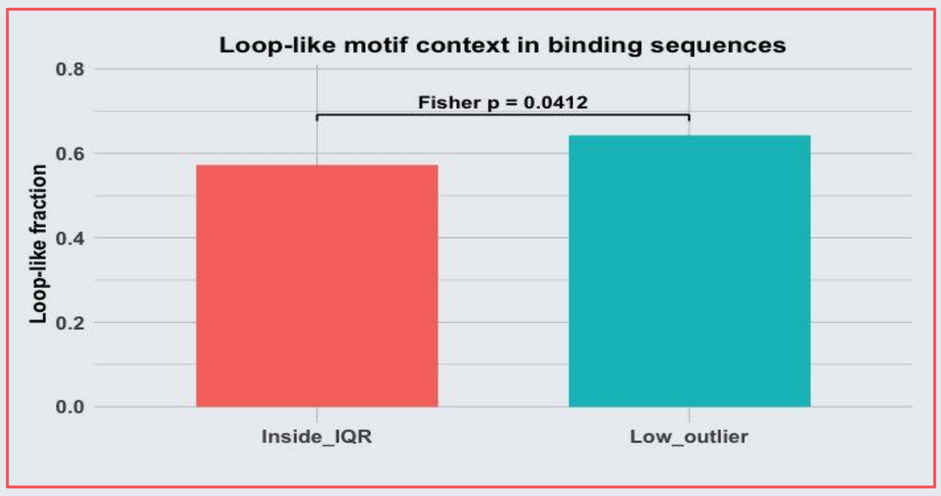
N_{Loop} – total number of binding/background sequences.

$(n_{binding} = 18,564 , n_{background} = 301)$

6. Example of binding sequences and their shuffled versions were used to visualize the structural context of representative motif variants.

Motif	UUAAUA (binding)	UUAAUA (background)	UUAUUU (binding)	UUAUUU (background)
ΔG $[\frac{Kcal}{mol}]$	- 6.2	- 5.3	- 4.3	- 3.9
Paired fraction	0.167	0.833	0.167	0.833

7. Binding sequences were divided based on ΔG, and loop-like motif occurrence was compared between central (IQR) and more stable (low ΔG) sequences to examine whether motif accessibility is preserved.



Conclusions

- Degenerate RNA motifs are enriched in Roquin-1 binding sites.
- Binding sequences exhibit lower predicted ΔG values, indicating increased global structural stability.
- Motifs preferentially localize to loop-like RNA regions.
- Loop localization is preserved even in highly stable binding sequences.
- The low appearance of the canonical CDE (UCYRYGA) suggests that Roquin-1 binding relies on AU-rich motifs in structured RNA, consistent with a dual mechanism combining ROQ-based structure recognition and Zn-finger-based sequence recognition.
- Roquin-1's flexible, degenerate binding enables a competitive advantage over sequence dependent RBPs and miRNAs in regulating mRNA stability.

References

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